Stopped Experiments, Bandit Convex Optimization, and Reinforcement Learning

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Abstract

We introduce stopped experiments for repeated decisions with a fixed unknown environment. Starting from a posterior whose Bayes risk is of order b, an adaptive experiment ends when the risk falls below γb, or when the experiment is abandoned. Its stopped cost is b times expected regret divided by the probability of this reduction. The posterior retains partial progress and determines how the experiment continues.

A uniform stopped bound SC gives $O(1+\sqrt{\mathsf{SC}T})$ Bayes regret, while a pointwise value c gives an $\Omega(\sqrt{cn})$ lower bound at a calibrated horizon. Consequently, the normalized-regret exponent lies between one half of the pointwise and uniform stopped exponents. Equality holds when the statewise experiments can be selected measurably. Finite-model episodic reinforcement learning fits by treating an episode policy as one decision and its trajectory as the observation.

For Borel classes of bounded continuous losses on compact action spaces with Gaussian feedback, measurable selection and Bayes–minimax equality give the exact exponent relation. In stochastic bandit convex optimization, arbitrary nonadaptive batches need size $\Omega(d^5)$, and stopped fresh-direction policies need expected duration $\Omega(d^5)$, to reduce risk with constant probability on a dimple family. An observation-adaptive experiment uses $\widetilde{O}(d^4)$ queries. Early observations are valuable because they orient later queries.

1 Introduction

How much regret must be spent before observations make a decision problem meaningfully easier? This question is not always answered by pricing one observation at a time. An early observation may reveal little by itself but determine where the next query should be made. The useful object is then the whole experiment: its observation-dependent continuation, the branches on which it is abandoned, and the first time at which the remaining decision problem becomes easier.

Stopped experiments. We begin with a fixed latent environment ω. An action a incurs a nonnegative decision gap $\Delta_\omega(a)$ and produces an observation. If Q_t is the posterior after t observations, its Bayes decision risk is

$$r(Q_t) = \inf_a \mathbb{E}_{Q_t}\Delta_\omega(a).$$

At a state where this risk is of order b, an adaptive experiment runs until it is abandoned or first reaches $r(Q_t) \le \gamma b$. If N is its duration and M the first-hit event, we define the pointwise stopped cost by

$$\mathsf{SC}_b(Q) = \inf_{\text{stopped experiments}} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}. \tag{1}$$

Thus $\mathsf{SC}_b(Q)$ is scaled regret per unit probability of a decision-relevant risk reduction. No dimension factor, random target, or information budget is included. A separate query charge is unnecessary: every query before the hit has expected gap at least γb, so $\mathbb{E}R_N \geq \gamma b\, \mathbb{E}N$.

A two-sided relation. Let $\mathsf{SC}^{\mathrm{pt}}(d, B)$ be the largest statewise value over risk scales $B^{-1} \leq b \leq 1$, and let $\mathsf{SC}^{\mathrm{unif}}(d, B)$ be the smallest bound attained by one measurable state-dependent experiment rule. The finite-horizon bounds have the form

$$\mathfrak{R}^{\mathrm{B}}(d, T) \lesssim_\gamma 1 + \sqrt{\mathsf{SC}^{\mathrm{unif}}(d, T)T}, \tag{2}$$

$$\mathfrak{R}^{\mathrm{B}}(d, n) \gtrsim_\gamma \sqrt{cn} \qquad \text{when } n \asymp_\gamma c/b^2, \tag{3}$$

for every $c < \mathsf{SC}^{\mathrm{pt}}(d, B)$, at some initial posterior and $b \in [B^{-1}, 1]$. The upper bound joins stopped experiments over geometric risk scales. For the lower bound, a policy that has not yet reduced risk pays at least γb per round, whereas reaching the lower-risk set with probability p costs at least cp/b. Balancing these two possibilities gives (3).

In terms of polynomial dimension exponents, the same theorem gives

$$\frac{\alpha^{\mathrm{pt}}_{\mathrm{stop}}}{2} \leq \alpha^{\mathrm{B}}_{\mathrm{reg}} \leq \frac{\alpha^{\mathrm{unif}}_{\mathrm{stop}}}{2}. \tag{4}$$

If the statewise experiments can be chosen measurably, the two stopped exponents agree and both inequalities are equalities. If Bayes and minimax regret also agree, then

$$\alpha^{\mathrm{M}}_{\mathrm{reg}} = \alpha^{\mathrm{B}}_{\mathrm{reg}} = \frac{\alpha_{\mathrm{stop}}}{2}.$$

These are exponent statements, not same-horizon equalities. They require nonnegative additive gaps, posterior sufficiency, and persistence of the same decision problem. They do not require convex losses or Gaussian observations.

Episodic reinforcement learning. Consider a fixed unknown finite-horizon MDP M that is reset from an initial distribution ρ at the beginning of every episode. Treat an entire within-episode policy π as the action and the resulting trajectory as its observation. The episode gap is

$$\Delta_M(\pi) = V^\star_M(\rho) - V^\pi_M(\rho) \geq 0.$$

The posterior risk

$$r(Q) = \inf_\pi \mathbb{E}_Q \Delta_M(\pi)$$

has exactly the persistence used above. Hence the stopped framework applies at the episode level. This macro-decision reduction includes standard episodic pseudo-regret. The finite-model corollary below gives automatic measurable selection and minimax closure when the model and trajectory alphabets are finite (Azar et al., 2017; Foster et al., 2021). Continuing RL is different. If the physical state is not reset, $r(Q_t, S_t)$ need not be a supermartingale: the next state may be harder even when no uncertainty is resolved. Such problems need an additional persistence condition or the full stateful framework of Xu (2026a).

Why bandit convex optimization? Stochastic bandit convex optimization (BCO) is a particularly revealing case. An unknown continuous convex loss $F : C \to [0, 1]$ on a compact convex body $C \subset \mathbb{R}^d$ is fixed. The learner chooses X_t, observes

$$O_t = F(X_t) + \xi_t, \qquad \xi_t \overset{\text{iid}}{\sim} N(0, 1),$$

and pays $F(X_t) - \min_C F$. The same scalar query is therefore both an experiment and a costly decision. Moreover, convex geometry links near-optimal regions to the locations and directions of informative probes, and the useful direction of a later probe may depend on earlier observations.

BCO also has a concrete unresolved dimension gap. In the nontrivial long-horizon regime, the best general upper bound is $\widetilde{O}(d^{3/2}\sqrt{T})$ (Lattimore and György, 2023; Fokkema et al., 2024). The recent lower bound $\widetilde{\Omega}(d^{5/4}\sqrt{T})$ of Rajaraman (2026) is the first full-class lower bound with dimension dependence strictly larger than the linear-bandit rate. Carpentier (2025) develops a complementary simple-regret formulation and a direct localization method for noisy zeroth-order convex optimization. Neither cumulative-regret endpoint is known to be sharp.

For every Borel class of bounded continuous losses on a compact action space under Gaussian feedback, not necessarily convex, we prove that the pointwise experiments can be chosen measurably and that worst-prior Bayes regret equals minimax regret. Thus

$$\alpha_{\text{reg}} = \frac{\alpha_{\text{stop}}}{2}$$

without an information-matching or nondegeneracy assumption. For the full convex class, the known regret bounds are equivalent to

$$\frac{5}{2} \leq \alpha_{\text{stop}} \leq 3.$$

Convexity is used here to define the BCO class, invoke its known bounds, and construct the geometric example below. It is not used in the stopped reduction.

Observation-adaptive continuation. For the dimple losses of Bakhtiari et al. (2025), at $\varepsilon = d^{-1}$ we prove

$$\mathsf{SC}_b^{[1]},\ \mathsf{SC}_b^{\text{fresh}} = \Omega(d^3), \qquad \mathsf{SC}_b = \widetilde{O}(d^2).$$

Every nonadaptive batch needs size $\Omega(d^5)$, and every stopped policy restricted to fresh directions—independent Haar directions with rays starting at the origin—needs expected duration $\Omega(d^5)$, for a constant-probability contraction. The unrestricted experiment first obtains a weak direction, uses tangent differences chosen from the observed direction to amplify it, and then contracts the angular error geometrically. It uses $\widetilde{O}(d^4)$ queries and yields $\widetilde{O}(\min\{T/d, d\sqrt{T}\})$ regret on this family. This is a separation between experiment classes, not a lower bound against all algorithms for the dimple family. Its role is to show that the stopped continuation can contain genuinely more geometry than its first query.

Information and statistical testing. The stopped cost is a decision cost, not an information measure. Mutual information, Le Cam, Fano, Assouad, and interactive change of measure remain useful for bounding the cost of reaching the lower-risk set; our fresh-direction lower bound uses mutual information in this way. Maximal information gain may likewise help bound $\mathsf{SC}_b(Q)$ in a particular model. It need not enter the definition, because the full posterior already records partial progress for the continuation of the experiment.

1.1 Related work

Blackwell comparison and the Bayesian theory of experiments evaluate observations through their effect on decision risk (Blackwell, 1953; Lindley, 1956; DeGroot, 1962). Classical sequential design optimizes both experiments and stopping (Chernoff, 1959; Naghshvar and Javidi, 2013). Here the stopping set is the first constant-factor reduction in Bayes decision risk, and the price is the regret incurred on the way to that set.

Information ratios, AIR, and DEC turn one-step regret–information or decision–estimation comparisons into sequential regret bounds (Russo and Van Roy, 2016, 2018; Lattimore and György, 2021; Foster et al., 2021, 2022, 2023; Xu and Zeevi, 2025; Neu et al., 2024). Chained ratios organize information across action scales (Gouverneur et al., 2024). These methods may re-optimize after every observation. Our distinction concerns the local object being optimized: the stopped cost keeps the continuation of one adaptive experiment rather than assigning a one-query value. Interactive Le Cam, Fano, and Assouad inequalities handle adaptive transcript laws directly (Chen et al., 2024).

Near-optimal targets and localization appear in rate–distortion, satisficing, and scale-sensitive ratio analyses (Dong and Van Roy, 2018; Russo and Van Roy, 2022; Arumugam and Van Roy, 2021; Bubeck and Sellke, 2023). A distinct modern line develops data-dependent localization through pointwise complexities (Li and Xu, 2026; Xu, 2026c,b). There, "pointwise" refers to localization around a realized input, distribution, or geometric point. Here it means that the best stopped experiment may be chosen separately at each posterior state; the implementation question is whether those choices form one measurable rule. The two notions are complementary. Polynomial gains from observations that orient later probes also arise in structured best-arm identification (Maiti et al., 2026).

The Bellman-sufficient framework of Xu (2026a) is broader: it allows physical states, contexts, and other history-dependent variables, and links same-horizon upper and lower bounds through a chosen information index. Its two sides meet when the corresponding information scales agree. The present paper studies the narrower repeated-decision case in which the posterior is sufficient and the nonnegative gap persists from one round to the next. Here the same stopped cost enters both directions, but the lower bound is evaluated at a horizon calibrated to the risk scale. On posterior space, the stopped experiment is a stochastic shortest-path problem (Bertsekas and Tsitsiklis, 1991); the new part is its two-sided relation to cumulative regret.

2 Stopped experiments for repeated decisions

2.1 The repeated latent decision problem

We first work beyond convex optimization, but within a repeated decision problem with a fixed latent environment. This is the static-state specialization of the Bellman-sufficient framework of Xu (2026a): the pre-action state is the current posterior, the same action problem recurs, and the action gap below is the nonnegative one-step Bellman loss. General stateful control may also admit a Bellman-sufficient state, but it need not have the persistence property used to join successive risk scales.

Let Ω, $\mathcal{A}$, and $\mathcal{O}$ be standard Borel environment, action, and observation spaces, and let $\Delta_\omega(a) \in [0, 1]$ be a nonnegative action gap. Querying a produces an observation in $\mathcal{O}$ from a kernel $P_\omega(\cdot \mid a)$. The environment $\omega \sim Q$ is drawn once and remains fixed. A policy chooses A_t from the pre-action history H_{t-1}, observes O_t, and incurs $\Delta_\omega(A_t)$. We include all independent exogenous policy randomization in H_0 and write $\mathcal{H}_t = \sigma(H_t)$ for the completed history filtration.

For a posterior Q, define the Bayes risk

$$r(Q) = \inf_{a \in \mathcal{A}} \mathbb{E}_Q \Delta_\omega(a).$$

We assume the posterior is a sufficient controlled state, r is measurable, and measurable ϵ-Bayes actions exist. These are standard in compact Borel models and are verified below for compact Gaussian loss classes. The theorem applies to static latent-model decision problems of the DEC/DMSO type. Neither the action space nor the map $a \mapsto \Delta_\omega(a)$ is assumed to be convex. The nonnegative gap assumption is substantive: with signed one-step increments, stopping can remove compensating future terms and elapsed time need not be paid by regret. Such problems require a separate query charge or a lower-bounded Bellman loss.

The repeated-decision structure implies that posterior Bayes risk is a nonnegative supermartingale. Indeed,

$$\mathbb{E}[r(Q_{t+1}) \mid \mathcal{H}_t] \leq \inf_{a \in \mathcal{A}} \mathbb{E}[\mathbb{E}\{\Delta_\omega(a) \mid \mathcal{H}_{t+1}\} \mid \mathcal{H}_t] = r(Q_t). \tag{5}$$

This fact, rather than convexity or a particular observation law, controls returns to a risk scale in the upper bound.

Example 2.1 (Episodic reinforcement learning)**.** *Let M be a fixed unknown MDP with episode length H, rewards in $[0,1]$, and a common initial-state distribution ρ. One macro-action is a complete contingent policy π for an episode, and its observation is the resulting trajectory. After normalizing by H, put*

$$\Delta_M(\pi) = \frac{V_M^\star(\rho) - V_M^\pi(\rho)}{H} \in [0,1].$$

The framework then measures episode pseudo-regret; a realized reward difference need not be nonnegative. At episode boundaries,

$$r(Q) = \inf_\pi \mathbb{E}_Q \Delta_M(\pi)$$

satisfies (5). For a finite model class, finite state, action, and reward alphabets, and a finite episode horizon, the required measurable choices and finite-horizon minimax equality reduce to finite-dimensional statements. Stopping takes place between episodes.

This reduction relies on the reset to ρ. In continuing control, the physical state changes and $r(Q_t, S_t)$ need not be a supermartingale. The next state can be intrinsically harder even without new uncertainty. One must then assume the corresponding risk-persistence inequality or use a stateful Bellman formulation such as Xu (2026a).

Fix

$$0 < \gamma < \vartheta < 1. \tag{6}$$

A pair $s = (Q, b)$ is *active* when

$$\vartheta b < r(Q) \leq b.$$

Starting from an active state, let

$$\tau_b = \inf\{t \geq 1 : r(Q_t) \leq \gamma b\} \tag{7}$$

be the first posterior-risk contraction.

Definition 2.2 (Stopped cost)**.** *An admissible experiment consists of an adaptive query policy and a positive, possibly infinite, $(\mathcal{H}_t)$-stopping time σ. It runs for*

$$N = \sigma \wedge \tau_b$$

rounds and succeeds on $\mathsf{M} = \{\tau_b \le \sigma\}$. *We require* $N < \infty$ *almost surely. Expectations below are initially extended-valued; Lemma 2.3 shows that every experiment with finite displayed ratio has* $\mathbb{E}N < \infty$. *Define*

$$\mathsf{SC}_b(Q) := \inf_{\pi,\sigma} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}, \qquad R_N = \sum_{t=1}^{N} \Delta_\omega(A_t), \tag{8}$$

with ratio $+\infty$ *when the success probability is zero.*

The first-hit convention loses nothing: after reaching the lower-risk set, additional observations cannot improve the success event and only add nonnegative regret. Voluntary stopping is essential, since an experiment may abandon an unpromising branch.

Lemma 2.3 (Query cost is already charged)**.** *Every admissible experiment satisfies*

$$\mathbb{E}_Q R_N \ge \gamma b\,\mathbb{E}_Q N. \tag{9}$$

Consequently,

$$\mathsf{SC}_b(Q) \le \inf_{\pi,\sigma} \frac{b^2\mathbb{E}_Q N + b\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})} \le (1+\gamma^{-1})\mathsf{SC}_b(Q).$$

Proof. The event $\{t \le N\}$ lies in $\mathcal{H}_{t-1}$, and on this event the pre-query posterior has not hit the lower-risk set. If π_t is the randomized action kernel, then

$$\mathbb{E}[\Delta_\omega(A_t) \mid \mathcal{H}_{t-1}] = \int \mathbb{E}_{Q_{t-1}} \Delta_\omega(a)\,\pi_t(da \mid H_{t-1}) \ge r(Q_{t-1}) > \gamma b.$$

Tonelli's theorem applied to the stopped sum proves (9); the comparison follows immediately. $\square$

2.2 The Bellman form of a stopped experiment

The stopped cost is a stochastic shortest-path problem on posterior space. For a query cap L and price $q > 0$, put $W_0^q = 0$ and

$$\begin{aligned} J_k^q(Q,b) = \inf_{a\in\mathcal{A}} \Big\{ & b\,\mathbb{E}_Q \Delta_\omega(a) \\ & + \int \big[-q\,\mathbf{1}\{r(Q^{a,o}) \le \gamma b\} + \mathbf{1}\{r(Q^{a,o}) > \gamma b\} W_{k-1}^q(Q^{a,o}, b)\big]\, P_Q(do \mid a) \Big\}, \\ W_k^q(Q,b) = \min\{0, & J_k^q(Q,b)\}. \end{aligned} \tag{10}$$

The root uses J_L^q, so the experiment makes at least one query; W_k^q permits voluntary stopping on a non-hitting branch. If $\mathsf{SC}_{b,L}$ is the infimum in (8) over experiments with $N \le L$ almost surely, then

$$\mathsf{SC}_{b,L}(Q) < q \iff J_L^q(Q,b) < 0, \qquad \mathsf{SC}_b(Q) = \inf_{L\ge1} \mathsf{SC}_{b,L}(Q). \tag{11}$$

The first equivalence follows by subtracting $q\mathbb{P}(\mathsf{M})$ from the stopped cost; the second follows by monotone truncation. Early observations receive value through $Q^{a,o}$ and its continuation, even when the first query does not itself hit.

2.3 Pointwise and uniform values

Let $\mathfrak{D} = (\mathfrak{D}_d)$ be a sequence of such decision problems. For a scale window $B^{-1} \le b \le 1$, write $\mathsf{S}_{\mathfrak{D}}(d, B)$ for the active posterior states. Define

$$\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d, B) = \sup_{(Q,b)\in \mathsf{S}_{\mathfrak{D}}(d,B)} \inf_{\mathcal{E}} \frac{b\, \mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}, \tag{12}$$

$$\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, B) = \inf_{\Sigma \text{ universally measurable}} \sup_{(Q,b)\in \mathsf{S}_{\mathfrak{D}}(d,B)} \frac{b\, \mathbb{E}^{\Sigma(Q,b)}_Q R_N}{\mathbb{P}^{\Sigma(Q,b)}_Q(\mathsf{M})}. \tag{13}$$

The supremum of an empty set is zero. Always $\mathsf{SC}^{\mathrm{pt}} \le \mathsf{SC}^{\mathrm{unif}}$. The first quantity chooses the best experiment separately at each state. The second requires all statewise choices to form one universally measurable experiment rule.

Let

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, T) = \sup_Q \inf_\pi \mathbb{E}^\pi_Q R_T$$

denote worst-prior Bayes regret, with any outer supremum over known action spaces understood.

Theorem 2.4 (Regret bounds from stopped experiments)**.** *For every $T \ge 2$ and $T^{-1} \le \eta < 1/2$,*

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, T) \le \min\left\{T,\, C_{\gamma,\vartheta} \frac{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, T)}{\eta} + 2T\eta\right\}. \tag{14}$$

In particular,

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, T) \le C_{\gamma,\vartheta} \min\left\{T,\, 1 + \sqrt{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, T)\, T}\right\}. \tag{15}$$

Conversely, for every $0 < c < \mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d, B)$, some active initial posterior and scale $B^{-1} \le b \le 1$ satisfy, for every $n \ge 1$,

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, n) \ge \frac{\gamma c b n}{c + \gamma b^2 n} \ge \frac{1}{2} \min\left\{\gamma b n,\, \frac{c}{b}\right\}. \tag{16}$$

If $c/(\gamma b^2) \ge 2$, then at

$$n = \left\lfloor \frac{c}{\gamma b^2} \right\rfloor$$

one has

$$\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, n) \ge \frac{\sqrt{\gamma}}{4} \sqrt{cn}. \tag{17}$$

Proof. If $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, T) = +\infty$, the upper statement is the trivial bound T. Otherwise fix one universally measurable selector Σ whose worst-state ratio is at most $\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, T) + \epsilon$. For the upper bound, use geometric risk buckets $b_j = \vartheta^j$. Whenever $r(Q_t) > 2\eta$, invoke $\Sigma(Q_t, b_j)$ at the unique active scale; otherwise play a measurable δ_t-Bayes action, where $\sum_t \delta_t \le \delta$. Virtually complete the last block if it crosses the horizon. Every actual block makes at least one query, so only this final block can cross T. At scale b_j, the selector's ratio bound gives

$$b_j\, \mathbb{E}[R_k \mid H_{S_k}] \le \{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, T) + \epsilon\} p_k,$$

where p_k is the conditional hit probability.

After the k-th hit in bucket j, at time U_k, one has $r(Q_{U_k}) \le \gamma b_j$. By (5) and Doob's maximal inequality, applied first over finite horizons after U_k and then by monotone convergence,

$$\mathbb{P}\left\{ \sup_{t \ge U_k} r(Q_t) > \vartheta b_j \,\middle|\, \mathcal{H}_{U_k} \right\} \le \frac{\gamma}{\vartheta}.$$

A later return to the same active bucket requires the event inside this probability. The number of hits in one bucket is therefore geometrically dominated and has expectation at most $(1 - \gamma/\vartheta)^{-1}$. Hence

$$\mathbb{E} R_{\text{explore}} \le \frac{\mathsf{SC}^{\text{unif}}_{\mathfrak{D}}(d, T) + \epsilon}{1 - \gamma/\vartheta} \sum_{b_j \ge 2\eta} \frac{1}{b_j} \le \frac{\mathsf{SC}^{\text{unif}}_{\mathfrak{D}}(d, T) + \epsilon}{2(1 - \vartheta)(1 - \gamma/\vartheta)\eta}.$$

The non-experiment rounds contribute at most $2T\eta + \delta$. Letting $\delta, \epsilon \downarrow 0$ proves (14); optimizing η, with the boundary choices and the trivial bound T, proves (15).

For the converse, choose an active state with $\mathsf{SC}_b(Q) > c$. Run an arbitrary n-round policy, stop it at $N = \tau_b \wedge n$, and put $p = \mathbb{P}_Q(\tau_b \le n)$. The stopped-cost premise and nonnegativity give

$$\mathbb{E}_Q R_n \ge \mathbb{E}_Q R_N \ge \frac{c}{b}\, p.$$

By the same pre-hit argument as in Lemma 2.3,

$$\mathbb{E}_Q R_n \ge \mathbb{E}_Q R_N \ge \gamma b \, \mathbb{E}_Q N \ge \gamma b n(1 - p).$$

Minimizing the maximum of these two affine functions over $p \in [0, 1]$ gives (16). At the displayed calibrated horizon, $n \ge c/(2\gamma b^2)$, and direct substitution gives $\mathbb{E}_Q R_n \ge c/(4b) \ge (\sqrt{\gamma}/4)\sqrt{cn}$. $\square$

2.4 Polynomial exponents

The finite-value bounds compare different scales and, on the lower side, different horizons. Polynomial exponents remove this calibration while retaining the distinction between statewise and measurably implementable experiments. Define

$$\alpha^{\text{stop,pt}}_{\mathfrak{D}} = \sup_{m \in \mathbb{N}} \limsup_{d \to \infty} \frac{\log\{1 + \mathsf{SC}^{\text{pt}}_{\mathfrak{D}}(d, d^m)\}}{\log d}, \tag{18}$$

$$\alpha^{\text{stop,unif}}_{\mathfrak{D}} = \sup_{m \in \mathbb{N}} \limsup_{d \to \infty} \frac{\log\{1 + \mathsf{SC}^{\text{unif}}_{\mathfrak{D}}(d, d^m)\}}{\log d}, \tag{19}$$

$$\alpha^{\text{B}}_{\mathfrak{D}} = \sup_{M \in \mathbb{N}} \limsup_{d \to \infty} \sup_{2 \le T \le d^M} \frac{\log\{1 + \mathfrak{R}^{\text{B}}_{\mathfrak{D}}(d, T)/\sqrt{T}\}}{\log d}. \tag{20}$$

If a frequentist minimax value $\mathfrak{R}^{\text{M}}_{\mathfrak{D}}$ is defined, let $\alpha^{\text{M}}_{\mathfrak{D}}$ be given by the same display.

Theorem 2.5 (Regret and stopped exponents)**.** *Every sequence of repeated latent decision problems satisfying the assumptions of this section obeys*

$$\frac{\alpha^{\text{stop,pt}}_{\mathfrak{D}}}{2} \le \alpha^{\text{B}}_{\mathfrak{D}} \le \frac{\alpha^{\text{stop,unif}}_{\mathfrak{D}}}{2}. \tag{21}$$

Neither inequality requires convexity or a particular observation law.

Suppose, in addition, that for every finite d, B and every $\epsilon > 0$, the statewise experiments have a universally measurable choice whose ratio at every active (Q, b) is at most $\mathsf{SC}_b(Q) + \epsilon$. Then

$$\mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d, B) = \mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, B) =: \mathsf{SC}_{\mathfrak{D}}(d, B),$$

and

$$\alpha^{\mathrm{B}}_{\mathfrak{D}} = \frac{\alpha^{\mathrm{stop}}_{\mathfrak{D}}}{2}. \tag{22}$$

If finite-horizon Bayes–minimax equality also holds, then

$$\alpha^{\mathrm{M}}_{\mathfrak{D}} = \alpha^{\mathrm{B}}_{\mathfrak{D}} = \frac{\alpha^{\mathrm{stop}}_{\mathfrak{D}}}{2}. \tag{23}$$

Proof. For $T \le d^M$, Theorem 2.4 and monotonicity of the scale window give

$$\frac{\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, T)}{\sqrt{T}} \le C_{\gamma,\vartheta} \left\{ T^{-1/2} + \sqrt{\mathsf{SC}^{\mathrm{unif}}_{\mathfrak{D}}(d, d^M)} \right\}.$$

Taking exponents proves the upper inequality in (21).

For the reverse inequality, first suppose $\alpha^{\mathrm{stop,pt}}_{\mathfrak{D}} > 0$, and fix

$$0 < q < q' < \alpha^{\mathrm{stop,pt}}_{\mathfrak{D}}.$$

For some fixed m and an infinite sequence of dimensions,

$$1 + \mathsf{SC}^{\mathrm{pt}}_{\mathfrak{D}}(d, d^m) \ge d^{q'}.$$

For all sufficiently large dimensions in this sequence, there is an active state (Q_d, b_d), with $b_d \ge d^{-m}$, such that

$$\mathsf{SC}_{b_d}(Q_d) > c_d, \qquad c_d = d^q.$$

Set

$$n_d = \left\lfloor \frac{c_d}{\gamma b_d^2} \right\rfloor.$$

Then $c_d/(\gamma b_d^2) \ge 2$ for large d, and

$$n_d \le \gamma^{-1} d^{q+2m} \le d^M$$

for one fixed integer M. The calibrated lower bound in Theorem 2.4 yields

$$\frac{\mathfrak{R}^{\mathrm{B}}_{\mathfrak{D}}(d, n_d)}{\sqrt{n_d}} \ge \frac{\sqrt{\gamma}}{4}\sqrt{c_d} = \frac{\sqrt{\gamma}}{4} d^{q/2}.$$

Thus $\alpha^{\mathrm{B}}_{\mathfrak{D}} \ge q/2$. Let $q \uparrow \alpha^{\mathrm{stop,pt}}_{\mathfrak{D}}$; if the latter exponent is infinite, take arbitrary finite q. When it is zero, the lower inequality is immediate.

The measurable-choice assumption identifies the two finite stopped values, and hence their exponents. Bayes–minimax equality identifies the remaining regret exponents. □

The equality does not require information-scale matching or nondegeneracy; it still assumes nonnegative gaps, posterior sufficiency, recurrence of the same decision problem, and measurable selection.

Corollary 2.6 (Finite episodic models)**.** *Consider any sequence of reset episodic reinforcement-learning problems as in Example 2.1, with a finite model class, finite state, action, and reward alphabets, and a finite episode horizon. Using the normalized episode gap,*

$$\alpha_{\mathrm{RL}}^{\mathrm{M}} = \alpha_{\mathrm{RL}}^{\mathrm{B}} = \frac{\alpha_{\mathrm{RL}}^{\mathrm{stop}}}{2}. \tag{24}$$

For fixed H, normalization does not change the dimension exponent. If H grows with d, its dependence must be restored when translating back to unnormalized regret.

Proof. The complete contingent policy space and the trajectory space are finite. The posterior belongs to a finite-dimensional simplex, the capped stopped problems admit measurable statewise choices, and truncation removes the cap. At every finite outer horizon there are finitely many pure nonanticipating policies, and randomized policies generate their compact convex hull. The finite minimax theorem gives Bayes–minimax equality. Apply Theorem 2.5. □

3 Gaussian feedback on compact action spaces

We next give a broad setting in which the statewise stopped experiments form one measurable rule and Bayes regret equals minimax regret. Let $\mathcal{C}_d$ be a collection of compact metric action spaces. For $C \in \mathcal{C}_d$, let $\mathfrak{F}_d(C)$ be a Borel subset of $\mathcal{C}(C, [0,1])$, with the uniform topology. If f is fixed, querying $x \in C$ reveals

$$O = f(x) + \xi, \qquad \xi \sim N(0,1),$$

and incurs the gap $f(x) - \min_C f$. Neither C nor f is assumed to be convex in this section. Write

$$\mathfrak{R}_{\mathfrak{F}}^{\mathrm{M}}(d,T) = \sup_{C\in\mathcal{C}_d} \inf_{\pi} \sup_{f\in\mathfrak{F}_d(C)} \mathbb{E}_f^{\pi} R_T, \tag{25}$$

$$\mathfrak{R}_{\mathfrak{F}}^{\mathrm{B}}(d,T) = \sup_{C\in\mathcal{C}_d} \sup_{Q \text{ on } \mathfrak{F}_d(C)} \inf_{\pi} \mathbb{E}_Q^{\pi} R_T. \tag{26}$$

The values $\mathsf{SC}_{\mathfrak{F},C}^{\mathrm{pt}}$ and $\mathsf{SC}_{\mathfrak{F},C}^{\mathrm{unif}}$ are defined by (12)–(13); an outer supremum over $C \in \mathcal{C}_d$ gives $\mathsf{SC}_{\mathfrak{F}}^{\mathrm{pt}}$ and $\mathsf{SC}_{\mathfrak{F}}^{\mathrm{unif}}$.

Theorem 3.1 (Gaussian statewise selection and minimax equality)**.** *For every finite d, T, B,*

$$\mathfrak{R}_{\mathfrak{F}}^{\mathrm{M}}(d,T) = \mathfrak{R}_{\mathfrak{F}}^{\mathrm{B}}(d,T), \tag{27}$$

$$\mathsf{SC}_{\mathfrak{F},C}^{\mathrm{pt}}(d,B) = \mathsf{SC}_{\mathfrak{F},C}^{\mathrm{unif}}(d,B). \tag{28}$$

The supremum in (27) may be restricted to finitely supported priors. For every $\epsilon > 0$, the second equality is implemented by a universally measurable state-dependent experiment whose finite deterministic cap may depend on the state.

The first equality is a finite-horizon minimax theorem, not an asymptotic interchange. The second identifies the statewise and uniform stopped values. Convexity is absent: compactness and continuity give Bayes-action minimizers, lower-semianalytic dynamic programming gives universally measurable near-minimizers, and the positive Gaussian density gives a Borel posterior update. Write their common worst-space value as $\mathsf{SC}_{\mathfrak{F}}(d,B)$, and use the exponents from (18)–(20) with $\mathfrak{D} = \mathfrak{F}$.

Corollary 3.2 (Regret and stopped exponents under Gaussian feedback)**.** *For every sequence of the Gaussian loss models above,*

$$\alpha_{\mathfrak{F}}^{\mathrm{M}} = \alpha_{\mathfrak{F}}^{\mathrm{B}} = \frac{\alpha_{\mathfrak{F}}^{\mathrm{stop}}}{2}. \tag{29}$$

No information-scale matching or nondegeneracy assumption is required.

Proof. Apply Theorem 2.5 and (27)–(28). □

3.1 The full convex class

We now return to stochastic BCO: $C \subset \mathbb{R}^d$ is a compact convex body and $\mathfrak{F}_d(C)$ is the full class of continuous $[0,1]$-valued convex losses. Convexity first becomes essential here, through the model class and the BCO bounds used below.

Corollary 3.3 (Full-class exponents)**.** *For the full class of continuous* $[0,1]$*-valued convex losses,*

$$\frac{5}{2} \le \alpha_{\mathrm{full}}^{\mathrm{stop}} \le 3, \qquad \frac{5}{4} \le \alpha_{\mathrm{full}}^{\mathrm{reg}} \le \frac{3}{2}. \tag{30}$$

Determining either exponent determines the other.

Proof. The upper endpoint is the current general BCO upper bound (Lattimore and György, 2023; Fokkema et al., 2024); the lower endpoint is the minimax theorem of Rajaraman (2026). Its 1-Lipschitz hard family embeds in the present bounded unit-noise model by the affine rescaling $g = (f+1)/2$: from an observation $f(x) + \xi$, return $(f(x) + \xi + 1)/2 + \zeta$, where $\zeta \sim N(0, 3/4)$ is independent. The result is $g(x) + \xi'$ with $\xi' \sim N(0,1)$, and regret changes only by a factor two. Apply Theorem 3.2. □

4 Why continuation matters: the dimple family

Let $C = B_2^d(1)$. Fix

$$d^{-1} \le \varepsilon \le \varepsilon_0, \qquad s = \sqrt{1 + 2\varepsilon},$$

where $\varepsilon_0 > 0$ is a sufficiently small constant. For $\theta \in S_2^{d-1}(1)$, define

$$F_\theta(x) = \varepsilon + \frac{1}{2}\|x\|^2 - \frac{1}{2}\big(s - \|\theta - x\|\big)_+^2. \tag{31}$$

These functions are continuous, convex, and $[0,1]$-valued after reducing ε_0 if necessary (Bakhtiari et al., 2025, Lemma 10). Their unique minimizers are θ, with $F_\theta(\theta) = 0$. At the origin,

$$b_0 := F_\theta(0) = s - 1 = \frac{2\varepsilon}{\sqrt{1+2\varepsilon} + 1} \asymp \varepsilon. \tag{32}$$

Bakhtiari et al. (2025) use this family to exhibit an obstruction to standard one-step information-ratio analysis. We show that the same geometry admits a cheaper multistep experiment. The lower bound below uses mutual information to show that an unoriented ray reveals very little; the upper construction uses early observations to orient later tangent probes.

Use the rotation-invariant prior $\Theta \sim \mathrm{Unif}(S_2^{d-1}(1))$. Since $\nabla F_\theta(0) = -b_0\theta$, convexity gives

$$F_\theta(x) \ge b_0\{1 - \langle \theta, x\rangle\}.$$

The origin is therefore the prior Bayes action and $r(Q_0) = b_0$. Write

$$\tau_\gamma = \inf\{t \ge 1 : r(Q_t) \le \gamma b_0\}. \tag{33}$$

Definition 4.1 (Fresh directions)**.** *A policy uses* fresh directions *if, at round t, it first draws $B_t \sim \mathrm{Unif}(S_2^{d-1}(1))$, independently of (Θ, H_{t-1}), then chooses an arbitrary $\sigma(H_{t-1}, B_t)$-measurable radius $R_t \in [0,1]$ and queries $X_t = R_t B_t$.*

The radius may use the complete past. Only the direction must be fresh. Every ray starts at the origin. The comparison excludes shifted or paired probes, anisotropic directions, and reuse of a learned direction.

Theorem 4.2 (Fresh-direction information bound)**.** *For $d \geq 2$, there is a universal $C < \infty$ such that, for every posterior ν of Θ, every fresh $B \sim \mathrm{Unif}(S_2^{d-1}(1))$, every measurable $R = R(B) \in [0,1]$, and independent $\xi \sim N(0,1)$,*

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \leq C(\varepsilon^4 + d^{-4}). \tag{34}$$

Under the spherical prior, every nonadaptive batch $X_1, \ldots, X_n$, possibly randomized but independent of Θ and fixed before its observations, satisfies

$$I(\Theta; O_{1:n} \mid X_{1:n}) \leq Cn(\varepsilon^4 + d^{-4}). \tag{35}$$

For any initial prior, every stopped fresh-direction experiment with duration N and $\mathbb{E}N < \infty$ satisfies

$$I(\Theta; H_N) \leq C(\varepsilon^4 + d^{-4})\, \mathbb{E}N. \tag{36}$$

Conversely, under the spherical initial prior, if $\mathsf{M} \in \sigma(H_N)$ is an event on which the terminal posterior Bayes risk is at most γb_0, then

$$I(\Theta; H_N) \geq \frac{d-1}{2}(1-\gamma)^2\, \mathbb{P}(\mathsf{M}). \tag{37}$$

Consequently, every stopped fresh-direction experiment under the spherical prior with $\mathbb{P}(\mathsf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathsf{M})} \geq c_\gamma \frac{d}{\varepsilon^4 + d^{-4}}. \tag{38}$$

The same bound holds with $N = n$ for an arbitrary nonadaptive batch of size n. Thus constant-probability contraction requires batch size, or expected fresh-direction duration, $\Omega_\gamma(d/(\varepsilon^4 + d^{-4}))$.

Fix a sufficiently large $C_L = C_L(\gamma)$, and for $0 < a \leq b_0$ put

$$L_{d,a} = 1 + \log\left\{\frac{C_L d(1 + \log(d/a))}{a}\right\}. \tag{39}$$

Theorem 4.3 (Observation-adaptive directional experiment)**.** *There are constants $\gamma_0, c_0 > 0$, with $\gamma_0 + c_0 < \gamma$, and $C_\gamma < \infty$ such that, for every sufficiently large d, every $d^{-1} \leq \varepsilon \leq \varepsilon_0$, and every $0 < a \leq b_0$, an observation-adaptive experiment returns $\widehat{x} \in B_2^d(1)$ with*

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{P}_\theta\{F_\theta(\widehat{x}) > \gamma_0 a\} \leq c_0 a, \tag{40}$$

$$N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a^2}, \tag{41}$$

$$\sup_{\theta \in S_2^{d-1}(1)} \mathbb{E}_\theta R_N \leq \frac{C_\gamma d^2 L_{d,a}^2}{a}. \tag{42}$$

The query cap is deterministic. The conditional distributions and subspaces of the directions in the last two stages depend measurably on earlier observations. Under every prior on Θ,

$$\mathbb{P}\left\{\mathbb{E}[F_\Theta(\widehat{x}) \mid H_N] \leq \gamma a\right\} \geq 1 - \frac{\gamma_0 + c_0}{\gamma} > 0. \tag{43}$$

The construction has three stages. The table suppresses logarithmic factors.

stage	progress	(queries, regret)
orthogonal scan	obtain v_0 with $\langle\theta, v_0\rangle = \widetilde{\Omega}(\max\{\sqrt{\varepsilon}, d^{-1/2}\})$	$(d^2/\varepsilon^2,\ d^2/\varepsilon)$
tangent amplification	choose tangent directions from the current estimate and enter a fixed angular cone	$(d^2/\varepsilon^2,\ d^2/\varepsilon)$
angular contraction	invert boundary observations and repeat $q \mapsto q/2$ until $q = O(a)$	$(d^2/a^2,\ d^2/a)$

Only the last two stages orient later probes using earlier observations. Their detailed analysis is in Appendix C.

For the next statement, $\mathsf{SC}_b^{[1]}$ restricts (8) to one query, and $\mathsf{SC}_b^{\text{fresh}}$ restricts it to fresh-direction policies.

Corollary 4.4 (Polynomial separations)**.** *For all sufficiently large d, at $\varepsilon = d^{-1}$ and $b = b_0 \asymp d^{-1}$, the initial spherical state satisfies*

$$\mathsf{SC}_b^{[1]} = \Omega_\gamma(d^3), \qquad \mathsf{SC}_b^{\text{fresh}} = \Omega_\gamma(d^3), \qquad \mathsf{SC}_b = \widetilde{O}_\gamma(d^2). \tag{44}$$

At the same scale, a nonadaptive batch with constant contraction probability has size $\Omega_\gamma(d^5)$, and every stopped fresh-direction experiment with $\mathbb{P}(\mathsf{M}) > 0$ satisfies

$$\frac{\mathbb{E}N}{\mathbb{P}(\mathsf{M})} \geq c_\gamma d^5.$$

In particular, constant success probability requires $\mathbb{E}N = \Omega_\gamma(d^5)$. The observation-adaptive construction contracts risk with constant probability using $\widetilde{O}_\gamma(d^4)$ queries.

Let

$$L_{d,T} = 1 + \log\left\{C_L d \max\{T, d\}[1 + \log\{d \max\{T, d\}\}]\right\}.$$

Under the spherical prior, every fresh-direction policy satisfies

$$\mathbb{E}R_T \geq c_\gamma \min\left\{\frac{T}{d}, d^4\right\}, \tag{45}$$

while an unrestricted policy satisfies, uniformly in θ,

$$\mathbb{E}_\theta R_T \leq C_\gamma L_{d,T} \min\left\{\frac{T}{d}, d\sqrt{T}\right\}. \tag{46}$$

At $T = d^5$, these bounds are $\Omega(d^4)$ and $\widetilde{O}(d^{7/2})$, respectively.

Proof of (44). For a one-query or fresh-direction first-hit experiment, let $p = \mathbb{P}(\mathsf{M})$. Theorem 4.2 gives

$$C(\varepsilon^4 + d^{-4})\,\mathbb{E}N \geq I(\Theta; H_N) \geq c_\gamma d\, p.$$

Before the hit, Lemma 2.3 gives $\mathbb{E}R_N \geq \gamma b\,\mathbb{E}N$. Therefore

$$\frac{b\,\mathbb{E}R_N}{p} \geq \frac{c_\gamma b^2 d}{\varepsilon^4 + d^{-4}}.$$

At $b \asymp \varepsilon = d^{-1}$, this is $\Omega_\gamma(d^3)$. The same argument covers the one-query restriction.

Couple the capped routine from Theorem 4.3, run at $a = b_0$, to its first-hit truncation. The two policies agree until the hit, and (43) implies that the virtual capped path hits with probability at least $p_\gamma := 1 - (\gamma_0 + c_0)/\gamma > 0$. Truncation can only decrease regret, so

$$\frac{b\,\mathbb{E}R_{\tau_\gamma \wedge N}}{\mathbb{P}(\tau_\gamma \leq N)} \leq \frac{C_\gamma}{p_\gamma} b \frac{d^2 L_{d,b}^2}{b} = \widetilde{O}_\gamma(d^2).$$

This proves the upper bound. $\square$

5 Discussion

The stopped cost asks how much regret must be spent before observations make a repeated decision problem easier. Bandit convex optimization shows why the multistep formulation is useful. On the dimple family, an unoriented ray carries little information about the hidden direction, but a weak initial direction can be reused to place tangent probes, amplify alignment, and contract angular error. This gives a polynomial separation between nonadaptive or fresh-direction experiments and an observation-adaptive experiment. It does not give a lower bound for unrestricted BCO.

For the full convex class, determining $\alpha_{\mathrm{full}}^{\mathrm{stop}} \in [5/2, 3]$ is equivalent, at the level of polynomial dimension exponents, to closing the current BCO regret gap. An upper bound must construct low-cost posterior experiments uniformly over states and scales; a lower bound must rule out their full observation-dependent continuation.

A The fresh-direction lower bound

Proof of Theorem 4.2. Put

$$q(x) = \varepsilon + \frac{1}{2}\|x\|^2.$$

For $B \in S_2^{d-1}(1)$, $r \in [0,1]$, and $u = \langle \theta, B \rangle$,

$$|F_\theta(rB) - q(rB)| = \frac{1}{2}\left(s - \sqrt{1 + r^2 - 2ru}\right)_+^2.$$

Minimizing the square root over $r \in [0,1]$ shows that the supremum of the right side is $(s-1)^2/2$ when $u \le 0$, and

$$\frac{1}{2}\left(s - \sqrt{1-u^2}\right)^2$$

when $u > 0$. On $|u| \le 1/2$, its square is at most $C(\varepsilon^4 + u^8)$; the complementary spherical cap has exponentially small probability. Rotation invariance and

$$\mathbb{E}_B \langle \theta, B \rangle^8 = \frac{105}{d(d+2)(d+4)(d+6)}$$

therefore give, uniformly in θ,

$$\mathbb{E}_B \sup_{0 \le r \le 1} |F_\theta(rB) - q(rB)|^2 \le C(\varepsilon^4 + d^{-4}). \tag{47}$$

Now condition on an arbitrary posterior ν. The fresh Haar direction B is independent of Θ, and $R(B)$ is known after conditioning on B. Comparing the Gaussian observation channel with the Θ-independent reference $q(R(B)B) + \xi$, the information-radius bound gives

$$I_\nu(\Theta; B, F_\Theta(RB) + \xi) \le \frac{1}{2}\mathbb{E}_{\Theta \sim \nu, B}|F_\Theta(RB) - q(RB)|^2.$$

Equation (47) proves (34).

For a fixed nonadaptive design point $x = ru$, rotation invariance interchanges a fixed direction u under uniform Θ with a uniform direction B under fixed θ. Thus (47) also gives

$$\mathbb{E}_\Theta |F_\Theta(x) - q(x)|^2 \le C(\varepsilon^4 + d^{-4}).$$

Condition on a possibly randomized design and compare its joint Gaussian observation law with the product reference $(q(X_i) + \xi_i)_{i \le n}$. Additivity of the reference KL divergence proves (35).

For a stopped fresh-direction experiment, let $N_n = N \wedge n$ and use the cemetery-padded transcript. Since $\{N \ge t\}$ is H_{t-1}-measurable, conditional application of (34) and the chain rule give

$$I(\Theta; H_{N_n}) \le C(\varepsilon^4 + d^{-4}) \sum_{t=1}^{n} \mathbb{P}(N \ge t) = C(\varepsilon^4 + d^{-4})\, \mathbb{E}N_n.$$

Continuity of relative entropy along the increasing stopped σ-fields, followed by $n \to \infty$, proves (36).

For the converse, convexity at the origin and $\nabla F_\theta(0) = -b_0\theta$ give

$$F_\theta(x) \ge b_0\{1 - \langle \theta, x \rangle\}. \tag{48}$$

With $m_N = \mathbb{E}[\Theta \mid H_N]$, taking the infimum over x in (48) gives directly

$$r(Q_{H_N}) \geq b_0(1 - \|m_N\|).$$

Hence $r(Q_{H_N}) \leq \gamma b_0$ on M implies $\|m_N\| \geq 1-\gamma$. The uniform law σ on $S_2^{d-1}(1)$ is $1/(d-1)$-sub-Gaussian. The variational formula for relative entropy therefore yields

$$D_{\mathrm{KL}}(\nu\|\sigma) \geq \frac{d-1}{2}\|\mathbb{E}_\nu \Theta\|^2.$$

Apply this to each posterior and average on M:

$$I(\Theta; H_N) = \mathbb{E} D_{\mathrm{KL}}(Q_{H_N}\|\sigma) \geq \frac{d-1}{2}(1-\gamma)^2 \mathbb{P}(\mathsf{M}).$$

Combining this inequality with (36) proves (38). □

Proof of the lower bound in Corollary 4.4. Let $N = \tau_\gamma \wedge T$ and $p = \mathbb{P}(\tau_\gamma \leq T)$. Before the contraction time, every action has conditional expected regret at least $r(Q_{t-1}) > \gamma b_0$. Hence

$$\mathbb{E} R_T \geq \gamma b_0 \, \mathbb{E} N. \tag{49}$$

Moreover,

$$\mathbb{E} N \geq T(1-p).$$

Apply Theorem 4.2 to the stopped transcript H_N. On $\{\tau_\gamma \leq T\}$, its terminal posterior risk is at most γb_0, and therefore

$$C d^{-4} \mathbb{E} N \geq I(\Theta; H_N) \geq c_\gamma d\, p$$

when $\varepsilon = d^{-1}$. Thus

$$\mathbb{E} N \geq c_\gamma d^5 p.$$

Minimizing $\max\{T(1-p), c_\gamma d^5 p\}$ over $p \in [0,1]$ gives $\mathbb{E} N \geq c_\gamma \min\{T, d^5\}$. Since $b_0 \asymp d^{-1}$, (49) proves (45). □

B The dimple regret bounds

Proof of the upper statements in Corollary 4.4. The same adaptive experiment, run to accuracy a and followed by commitment, gives the family upper bound. At $a = b_0 \asymp \varepsilon$, Theorem 4.3 uses $\widetilde{O}(d^2/\varepsilon^2)$ queries and $\widetilde{O}(d^2/\varepsilon)$ regret. Equation (43) proves the adaptive part of the query separation. Since $\varepsilon^4 + d^{-4} \leq 2\varepsilon^4$ in the stated range, Theorem 4.2 proves the lower part for both nonadaptive and fresh-direction experiments. Specializing to $\varepsilon = d^{-1}$ gives the displayed endpoint.

For the regret upper bound, first note that playing the origin for all T rounds costs $b_0 T \leq CT/d$. When $T \geq C_\gamma d^4 L_{d,T}^2$, choose

$$a = A_\gamma \frac{d L_{d,T}}{\sqrt{T}}$$

with A_γ large enough. Then $a \leq b_0$, $L_{d,a} \leq C L_{d,T}$, and (41) uses at most $T/2$ rounds. Indeed, enlarging the crossover constant makes $a \leq c/d \leq b_0$, while $1/a \leq \sqrt{T}/d$ and $\log(d/a) \leq \log(dT)$ give the displayed logarithmic comparison directly from (39). The exploration regret is at most

$$\frac{C_\gamma d^2 L_{d,a}^2}{a} \leq C_\gamma d L_{d,T}\sqrt{T}.$$

Play $\widehat{x}$ for the remaining rounds. Since losses are bounded by one, (40) gives

$$\sup_\theta \mathbb{E}_\theta F_\theta(\widehat{x}) \le (\gamma_0 + c_0)a \le C_\gamma a.$$

The exploitation regret is therefore at most $C_\gamma aT \le C_\gamma d L_{d,T}\sqrt{T}$. Below the displayed crossover, use the origin strategy. If $T \le d^4$, its cost CT/d is the smaller term in (46). If $d^4 < T < C_\gamma d^4 L_{d,T}^2$, then

$$\frac{T}{d} \le C_\gamma L_{d,T}\, d\sqrt{T}.$$

These two cases prove (46) throughout the remaining range. □

C Proof of the observation-adaptive experiment

Throughout this appendix, fix $0 < a \le b_0$ and total failure probability $\delta = c_\delta a$, where $c_\delta > 0$ is fixed below, and put

$$\mathcal{L} = 1 + \log\{16d(1 + \log(d/a))/\delta\}.$$

This agrees with $L_{d,a}$ after increasing C_γ. Fix a sufficiently large universal C_{geo}, then choose $q_\star > 0$ so that

$$q_\star < s/4, \qquad C_{\text{geo}} q_\star \le s/2$$

uniformly over the stated range of ε. Next choose $\gamma_0, c_\delta > 0$ so that $\gamma_0 + c_\delta < \gamma$, and set $c_0 = c_\delta$, followed by the small geometric steps, termination tolerances, and large repetition constants used below. All constants are fixed in this order.

We repeatedly use the standard consequence of sphere concentration that, for a Haar orthonormal basis (u_i) in a fixed k-dimensional subspace and a fixed unit vector w,

$$\max_i |\langle w, u_i\rangle| \le C\sqrt{\mathcal{L}/k} \tag{50}$$

outside probability $e^{-c\mathcal{L}}$ when $\mathcal{L} \le k$; when $\mathcal{L} > k$, we use the deterministic bound one. Every repetition below uses fresh Gaussian noise. Each stage has a deterministic query cap, the number of stages is at most $C(1 + \log(d/a))$, and confidence levels are divided across these caps so that the union of all concentration failures is at most δ.

Every use of Haar concentration is conditional on the transcript before the corresponding basis is drawn. At that point, the current iterate, the hidden direction, and the activity margin are fixed, while the new basis is Haar in its tangent space. Iterated conditioning and the allocated union bound give a good event $\mathcal{H}$ on which all activity, estimation, and contraction invariants below hold. The deterministic caps define the routine on $\mathcal{H}^c$. The procedure is made total and Borel by returning the origin whenever a normalized estimate is undefined and by resolving every gate equality with the stopping branch.

Whenever $r = \|\theta - x\| < s$, expanding (31) gives

$$F_\theta(x) = s\|\theta - x\| + \langle \theta, x\rangle - 1. \tag{51}$$

C.1 Orthogonal initialization

Lemma C.1. *With probability at least $1 - \delta/3$, the construction below returns a unit vector v_0 satisfying*

$$\langle \theta, v_0\rangle \ge c\min\left\{1, \max\left(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\right)\right\}. \tag{52}$$

It uses at most $Cd^2\mathcal{L}^2/\varepsilon^2$ queries and incurs at most $Cd^2\mathcal{L}^2/\varepsilon$ regret on the success event.

Proof. Draw a Haar basis $(u_i)_{i=1}^d$, and set

$$h_0 = c_h \min\{\sqrt{\varepsilon}, \varepsilon\sqrt{d/\mathcal{L}}\}, \qquad n_0 = \lceil Cd\mathcal{L}^2/\varepsilon^2 \rceil. \tag{53}$$

Estimate every centered difference

$$D_i^0 = \frac{F_\theta(h_0 u_i) - F_\theta(-h_0 u_i)}{2h_0}$$

using n_0 repetitions at each of the two points. On (50),

$$h_0^2 + 2h_0 \max_i |\langle \theta, u_i \rangle| < 2\varepsilon$$

for sufficiently small c_h, so both points lie in the active dimple. Equation (51) gives

$$D_i^0 = -\lambda_i \langle \theta, u_i \rangle, \qquad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1, \qquad |\lambda_i - b_0| \le Ch_0^2, \tag{54}$$

where $r_i^\pm = \|\theta \mp h_0 u_i\|$. Hence the noiseless difference vector is

$$D = -b_0\theta + e_0, \qquad \|e_0\| \le Ch_0^2.$$

The Gaussian estimation error ζ is isotropic with coordinate variance $(2n_0h_0^2)^{-1}$. At the allocated joint failure level,

$$|\langle \theta, \zeta \rangle| \le \frac{C\varepsilon}{h_0\sqrt{d\mathcal{L}}}, \tag{55}$$

$$\|\zeta\| \le \frac{C\varepsilon}{h_0\mathcal{L}}\left(1 + \sqrt{\mathcal{L}/d}\right). \tag{56}$$

Let

$$t = \min\{1, \max(\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d})\}.$$

When $\mathcal{L} \le d$, the definition of h_0 gives $h_0 \asymp c_h\varepsilon/t$. After choosing c_h small and then the repetition constant large,

$$\langle \theta, -\widehat{D} \rangle \ge c\varepsilon, \qquad \|\widehat{D}\| \le C\varepsilon/t.$$

When $\mathcal{L} > d$, one has $t = 1$; the same conclusions follow from (56), $\varepsilon \ge d^{-1}$, and the definition of h_0. Set

$$v_0 = -\widehat{D}/\|\widehat{D}\|.$$

Then (52) holds.

The stage uses $2dn_0 = O(d^2\mathcal{L}^2/\varepsilon^2)$ queries. Every query point has loss $O(\varepsilon)$, so its regret is $O(d^2\mathcal{L}^2/\varepsilon)$. □

C.2 Tangent amplification

Lemma C.2. *Conditionally on the initialization event, a capped tangent routine reaches $\|\theta - v\| \le q_\star$ with failure probability at most $\delta/3$. If c_{init} denotes the right side of (52), the stage uses*

$$N_{\text{amp}} \le \frac{Cd^2\mathcal{L}}{c_{\text{init}}^4}, \qquad R_{\text{amp}} \le \frac{Cd^2\mathcal{L}}{c_{\text{init}}^2}, \tag{57}$$

up to lower-order $C\mathcal{L}(1 + \log(1/c_{\text{init}}))$ gate costs.

Proof. Suppose a unit vector v satisfies $\langle \theta, v\rangle \geq c$, where

$$c_{\mathrm{init}} \leq c \leq 1 - q_\star^2/8.$$

The constants ensure $c^2 \geq c_1\varepsilon$. On the unit sphere, $F_\theta(v)$ is the nondecreasing function

$$\phi_\varepsilon(q) = \varepsilon + \frac{1}{2} - \frac{1}{2}(s-q)_+^2, \qquad q = \|\theta - v\|.$$

The fixed choice of $q_\star$ makes

$$\phi_\varepsilon(q_\star) - \phi_\varepsilon(q_\star/2)$$

uniformly positive. Averaging $O(\mathcal{L})$ repeated observations at v therefore separates $q \leq q_\star/2$ from $q \geq q_\star$, with either decision allowed in the overlap. Stopping implies $q \leq q_\star$, while continuation implies $q \geq q_\star/2$.

On a continuing stage, put $x = (c/2)v$ and draw a Haar tangent basis $(u_i)_{i=1}^{d-1}$ of $v^\perp$. Write

$$p = \langle \theta, v\rangle, \qquad r_0 = \|\theta - x\|.$$

Then

$$r_0^2 = 1 + \frac{c^2}{4} - cp \leq 1 - \frac{3c^2}{4}, \qquad r_0 \geq 1 - \frac{c}{2}. \tag{58}$$

When $\mathcal{L} \leq d$, take $h = c_2 c$. On $\mathcal{H}$, (50) and $c \geq c_{\mathrm{init}} \gtrsim \sqrt{\mathcal{L}/d}$ give

$$|\langle \theta, u_i\rangle| \leq Cc.$$

Hence $x \pm hu_i$ remain in the active dimple for small enough c_2. When $\mathcal{L} > d$, take $h = c_2c^2$ and use the deterministic bound one; here c is bounded below by a universal positive constant, so the same conclusions hold after changing constants.

Let $q_i = \langle \theta, u_i\rangle$. The exact centered differences are

$$D_i = -\lambda_i q_i, \qquad \lambda_i = \frac{2s}{r_i^+ + r_i^-} - 1. \tag{59}$$

Writing $\lambda_0 = s/r_0 - 1$, a second-order Taylor expansion of $\sqrt{r_0^2 + t}$ at $t_\pm = h^2 \mp 2hq_i$ gives

$$c_3c^2 \leq \lambda_0 \leq Cc, \qquad |\lambda_i - \lambda_0| \leq Ch^2.$$

Put

$$g = -\sum_i D_i u_i = \sum_i \lambda_i q_i u_i.$$

Because the continuation gate ensures $\|\theta - v\| \geq q_\star/2$, the tangent component of θ is bounded below. Choosing c_2 sufficiently small gives

$$\langle \theta, g\rangle \geq c_4 c^2, \qquad \|g\| \leq Cc. \tag{60}$$

Estimate every D_i with

$$n_c = \lceil Cd\mathcal{L}/c^4\rceil$$

repetitions at each side. Gaussian concentration in $v^\perp$ gives an error vector ζ satisfying

$$\|\zeta\| \leq c_{\mathrm{amp}}c, \qquad |\langle \theta, \zeta\rangle| \leq (c_4/2)c^2$$

on the joint event. More explicitly, with empirical differences $\widehat{D}_i$, set

$$\widehat{g} := -\sum_i \widehat{D}_i u_i = g + \zeta, \qquad u := \widehat{g}/\|\widehat{g}\|.$$

Then $u \in v^\perp$ and

$$\langle \theta, u \rangle \geq k_0 c$$

for a universal $k_0 > 0$. Choose a fixed $0 < k \leq \min\{k_0, q_\star/4\}$, and replace

$$v \quad \text{by} \quad \frac{v + ku}{\sqrt{1+k^2}}.$$

Its correlation lower bound increases from c to at least $\sqrt{1+k^2}\,c$. There are $O(\log(2/c_{\text{init}}))$ stages.

At correlation c, the stage uses $O(d^2\mathcal{L}/c^4)$ queries. Each query has loss $O(\varepsilon + c^2) = O(c^2)$. Geometric summation gives (57). Since

$$c_{\text{init}} \gtrsim \max\{\sqrt{\varepsilon}, \sqrt{\mathcal{L}/d}\}$$

unless it is already a fixed constant, these costs are at most

$$O(d^2\mathcal{L}/\varepsilon^2) \quad \text{and} \quad O(d^2\mathcal{L}/\varepsilon),$$

respectively. □

C.3 Angular contraction

Lemma C.3. *Conditionally on the first two stage events, the capped contraction stage returns $\widehat{x}$ with $F_\theta(\widehat{x}) \leq \gamma_0 a$ outside probability $\delta/3$. It has deterministic query cap*

$$Cd^2\mathcal{L}/a^2$$

and success-event regret at most

$$Cd^2\mathcal{L}/a.$$

Proof. Maintain the invariant

$$\|\theta - v_j\| \leq Q_j, \qquad Q_j = q_\star 2^{-j}.$$

At one step, write $v = v_j$, $Q = Q_j$, take $h = c_5 Q$, and draw a Haar tangent basis (u_i) of $v^\perp$. Define

$$\alpha_h = (1+h^2)^{-1/2}, \qquad t = h(1+h^2)^{-1/2}, \qquad x_i^\pm = \alpha_h v \pm t u_i.$$

These points lie on the unit sphere and within distance CQ of θ, hence in the active dimple.

Write

$$\theta = c_v v + \sum_i q_i u_i, \qquad c_v = 1 - \frac{1}{2}\|\theta - v\|^2,$$

and let

$$\psi(r) = sr - \frac{1}{2}r^2, \qquad G(p) = \psi(\sqrt{2-2p}).$$

Equation (51) gives the exact inversion

$$q_i = \frac{G^{-1}(F_\theta(x_i^+)) - G^{-1}(F_\theta(x_i^-))}{2t}. \tag{61}$$

All inner products in this display belong to

$$I_Q = \{p \in [-1, 1] : \sqrt{2 - 2p} \le C_{\mathrm{geo}} Q\}.$$

On this interval, G is strictly monotone and

$$|(G^{-1})'(G(p))| = \frac{\sqrt{2 - 2p}}{s - \sqrt{2 - 2p}} \le CQ.$$

The inverse derivative extends continuously to $p = 1$.

Project each empirical function value onto $G(I_Q)$ before applying G^{-1}. Estimate each value to accuracy $c_6 Q/\sqrt{d}$, using

$$\lceil C d\mathcal{L}/Q^2 \rceil$$

repetitions per site. Every recovered tangent coefficient in (61) then has error at most $Cc_6 Q/\sqrt{d}$. If

$$w = \sum_i q_i u_i, \qquad \widehat{w} = \sum_i \widehat{q}_i u_i,$$

then $\|\widehat{w} - w\| \le Cc_6 Q$. Clip $\widehat{w}$ to norm at most $1/2$, and set

$$v_{\mathrm{new}} = \sqrt{1 - \|\widehat{w}\|^2}\, v + \widehat{w}.$$

The map $w \mapsto \sqrt{1 - \|w\|^2}\, v + w$ is 2-Lipschitz on this ball. Choosing c_6 sufficiently small yields

$$\|\theta - v_{\mathrm{new}}\| \le Q/2.$$

At scale Q, the query count is

$$O(d^2\mathcal{L}/Q^2).$$

Every query has loss $\psi(O(Q)) = O(Q)$, so the regret is

$$O(d^2\mathcal{L}/Q).$$

Geometric summation until $Q \le c_{\mathrm{term}} a$ proves the displayed query and regret bounds. For sufficiently small c_{term},

$$F_\theta(\widehat{x}) = \psi(\|\theta - \widehat{x}\|) \le \gamma_0 a.$$

$\square$

Proof of Theorem 4.3. Run Lemmas C.1–C.3 in order, assigning failure probability $\delta/3$ to each. Because $a \le b_0 \asymp \varepsilon$, their deterministic caps and geometric sums give

$$N \le C_\gamma d^2 L_{d,a}^2/a^2$$

and, on $\mathcal{H}$,

$$R_N \le C_\gamma d^2 L_{d,a}^2/a.$$

On $\mathcal{H}^c$, the capped routine uses no more than its displayed query budget and incurs at most one unit of regret per query. Since $\mathbb{P}(\mathcal{H}^c) \le \delta = c_\delta a$, this adds at most

$$C_\gamma d^2 L_{d,a}^2/a$$

to expected regret. The accuracy statement follows from Lemma C.3 and the union bound. Since losses are bounded by one, the same statement gives, under every prior,

$$\mathbb{E}F_\Theta(\widehat{x}) \le (\gamma_0 + c_0)a.$$

Apply Markov's inequality to the nonnegative random variable $\mathbb{E}[F_\Theta(\widehat{x}) \mid H_N]$ to obtain (43). On the displayed event, the posterior Bayes risk is no larger than the posterior loss of $\widehat{x}$, and hence is at most γa. $\square$

D Gaussian models and Bayes–minimax equality

This appendix proves Theorem 3.1. It establishes measurable implementation and the order of the Bayes and minimax optimizations; the two-sided stopped reduction itself was proved in Section 2.

Lemma D.1 (Prior-wise Borel realization)**.** *Fix a Borel prior Q and a finite horizon T. On standard Borel history, action, and observation spaces, every universally measurable randomized nonanticipating policy has a Borel randomized nonanticipating policy inducing the same joint law of (F, H_T) under Q. The same holds when the action space is augmented by a stopping symbol.*

Proof. At time t, represent the action kernel as a universally measurable map from histories into the standard Borel space of action distributions. Under the reached history law, this map has a Borel version. Replace it by that version; the strategic law through time t is unchanged. Induction over $t = 1, \ldots, T$ proves the claim. Adding one isolated stopping symbol preserves the standard Borel property. $\square$

D.1 Finite-horizon Bayes–minimax identity

Proof of (27). Let $\nu = N(0, 1)$, let $\mathsf{H}_T = (C \times \mathbb{R})^T$, and let $\mathcal{P}_T$ be the set of reference strategic measures

$$P(dh_T) = \prod_{t=1}^{T} \pi_t(dx_t \mid h_{t-1})\, \nu(dy_t)$$

generated by Borel policies. The set $\mathcal{P}_T$ is convex and weakly compact. Indeed, C is compact and every member has $\nu^{\otimes T}$ as its observation marginal, so the family is tight. It is closed because the conditional-independence identities

$$\int a(h_{t-1}, x_t) b(y_t)\, dP = \left(\int b\, d\nu \right) \int a(h_{t-1}, x_t)\, dP$$

for bounded continuous a, b pass to weak limits. A monotone-class argument extends these identities to bounded measurable functions and recovers the conditional law $Y_t \mid (H_{t-1}, X_t) \sim \nu$. Standard-Borel disintegration then identifies each limiting strategic measure with a Borel nonanticipating policy.

For $f \in \mathfrak{F}_d(C)$, put

$$L_f(h_T) = \exp\left\{ \sum_{t=1}^{T} \left(y_t f(x_t) - \frac{1}{2} f(x_t)^2 \right) \right\}$$

and

$$u(P, f) = \int L_f(h_T) \sum_{t=1}^{T} \{ f(x_t) - \min_C f \}\, dP(h_T).$$

Gaussian change of measure gives $u(P, f) = \mathbb{E}_f^{\pi} R_T$. The payoff is affine and continuous in P, and continuous in f under the uniform topology. Continuity follows by domination with

$$T \exp\left\{ \sum_{t=1}^{T} |y_t| \right\},$$

which is integrable uniformly over $\mathcal{P}_T$.

Let $\mathcal{Q}_0$ be the convex set of finitely supported priors on $\mathfrak{F}_d(C)$, and set $u(P,Q) = \int u(P,f)Q(df)$. Sion's theorem (Sion, 1958), with compact side $\mathcal{P}_T$, yields

$$\min_{P\in\mathcal{P}_T} \sup_f u(P,f) = \sup_{Q\in\mathcal{Q}_0} \min_{P\in\mathcal{P}_T} u(P,Q).$$

Allowing all Borel priors cannot decrease the right-hand side, and weak duality prevents it from exceeding the left-hand side. Taking the outer supremum over C proves (27). □

D.2 Measurable selection of stopped experiments

Lemma D.2 (Capped Gaussian selection)**.** *Fix a query cap $L < \infty$. The capped stopped value*

$$v_L(Q,b) = \inf_{\mathcal{E}:N\le L} \frac{b\,\mathbb{E}_Q R_N}{\mathbb{P}_Q(\mathsf{M})}$$

is universally measurable. For every $\epsilon > 0$, there is a universally measurable state-to-experiment rule with value at most $v_L(Q,b)+\epsilon$ at every active state.

Proof. The belief space $\mathcal{P}(\mathfrak{F}_d(C))$ is standard Borel. For a belief μ, define

$$\ell(\mu,x) = \int \{f(x) - \min_C f\}\,\mu(df), \qquad r(\mu) = \min_{x\in C} \ell(\mu,x).$$

The first map is Borel and continuous in x; compactness of C makes r Borel and supplies a Borel Bayes-action selector. With φ the standard Gaussian density, the posterior update is the Borel map

$$\Psi(\mu,x,o)(df) = \frac{\varphi(o-f(x))\,\mu(df)}{\int \varphi(o-f'(x))\,\mu(df')}.$$

For a rational price $q > 0$, the capped penalized problem is exactly the recursion (10), with $\mu' = \Psi(\mu,x,o)$. Its one-step costs are bounded below and all primitive kernels are Borel. Standard integration, partial minimization, and universally measurable selection for lower-semianalytic dynamic programs therefore apply (Bertsekas and Shreve, 1978, Propositions 7.47–7.50). In particular, its root value $J_L^q(\mu,b)$ is lower semianalytic and has universally measurable near-minimizers.

Directly from the definition,

$$v_L(\mu,b) < q \quad\Longleftrightarrow\quad J_L^q(\mu,b) < 0.$$

The rational strict sublevel sets make v_L universally measurable. On $\{v_L < +\infty\}$, enumerate the positive rationals and measurably choose the first q with

$$v_L + \epsilon/4 < q < v_L + \epsilon/2.$$

Then $J_L^q < 0$. Partition this set further by the first integer n such that $J_L^q \le -1/n$, and use a Bellman selector whose approximation error is less than $1/(2n)$. The resulting penalized value remains negative, so its success probability is positive and its ratio is below $q < v_L + \epsilon/2$. On $\{v_L = +\infty\}$, any measurable one-query rule satisfies the extended-value approximation. Countable patching gives one universally measurable ϵ-optimal rule. □

Lemma D.3 (Removing the query cap)**.** *The unrestricted value $\mathsf{SC}_b(Q)$ is universally measurable and has a universally measurable ϵ-optimal selector for every $\epsilon > 0$. The selected experiment may have a finite deterministic cap $L(Q,b)$ depending on the state.*

Proof. Truncate an admissible experiment after L queries. Its stopped regret and hit probability increase to those of the complete experiment:

$$\mathbb{E}R_{N\wedge L} \uparrow \mathbb{E}R_N, \qquad \mathbb{P}(\mathsf{M}_L) \uparrow \mathbb{P}(\mathsf{M}).$$

Hence its ratio converges, including the zero-success case under the $+\infty$ convention. Indeed, if $\mathbb{P}(\mathsf{M}) > 0$, componentwise convergence gives convergence of the ratios; if $\mathbb{P}(\mathsf{M}) = 0$, then every truncated hit probability is zero and all ratios are $+\infty$. Since capped experiments are a subset of all experiments, $\mathsf{SC}_b(Q) \le v_L(Q, b)$ for every L. Conversely, truncating any admissible experiment and passing to the limit gives the reverse inequality. Therefore

$$\mathsf{SC}_b(Q) = \inf_{L\ge 1} v_L(Q, b).$$

This is universally measurable by Lemma D.2. On $\{\mathsf{SC}_b < +\infty\}$, choose measurably the first L for which $v_L < \mathsf{SC}_b + \epsilon/2$, then use the capped $\epsilon/2$-selector. On $\{\mathsf{SC}_b = +\infty\}$, use any measurable one-query rule; the approximation statement is vacuous there. Countable patching proves the claim. □

Proof of (28). Lemma D.3 selects, simultaneously at every active state, an experiment within ϵ of its pointwise infimum. Taking the worst state and then letting $\epsilon \downarrow 0$ gives $\mathsf{SC}^{\mathrm{unif}} \le \mathsf{SC}^{\mathrm{pt}}$. The reverse inequality is immediate from the definitions. For each fixed prior and finite global horizon, Lemma D.1 replaces the universal selector by a Borel policy with the same strategic law. □

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